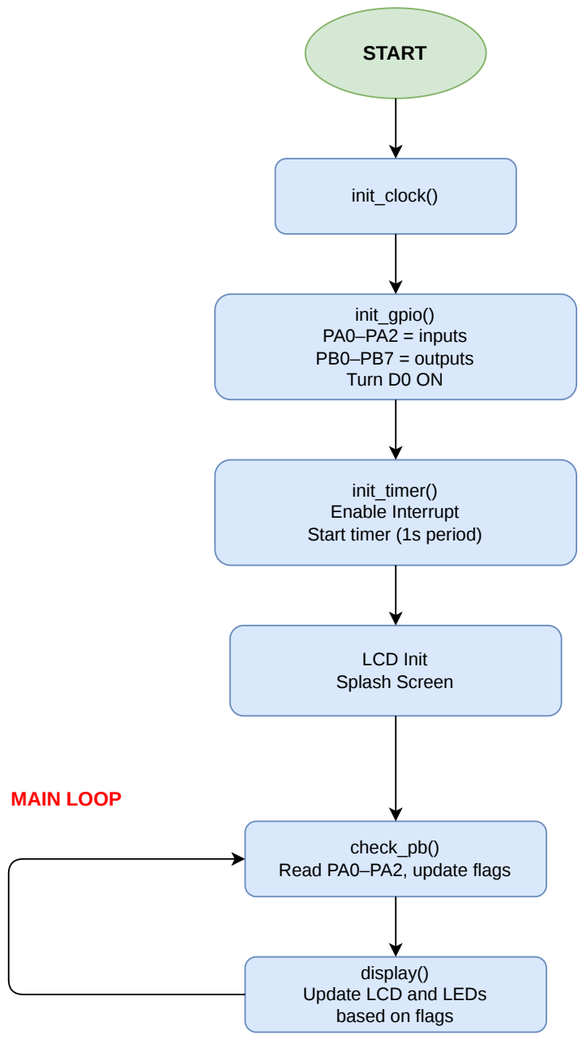
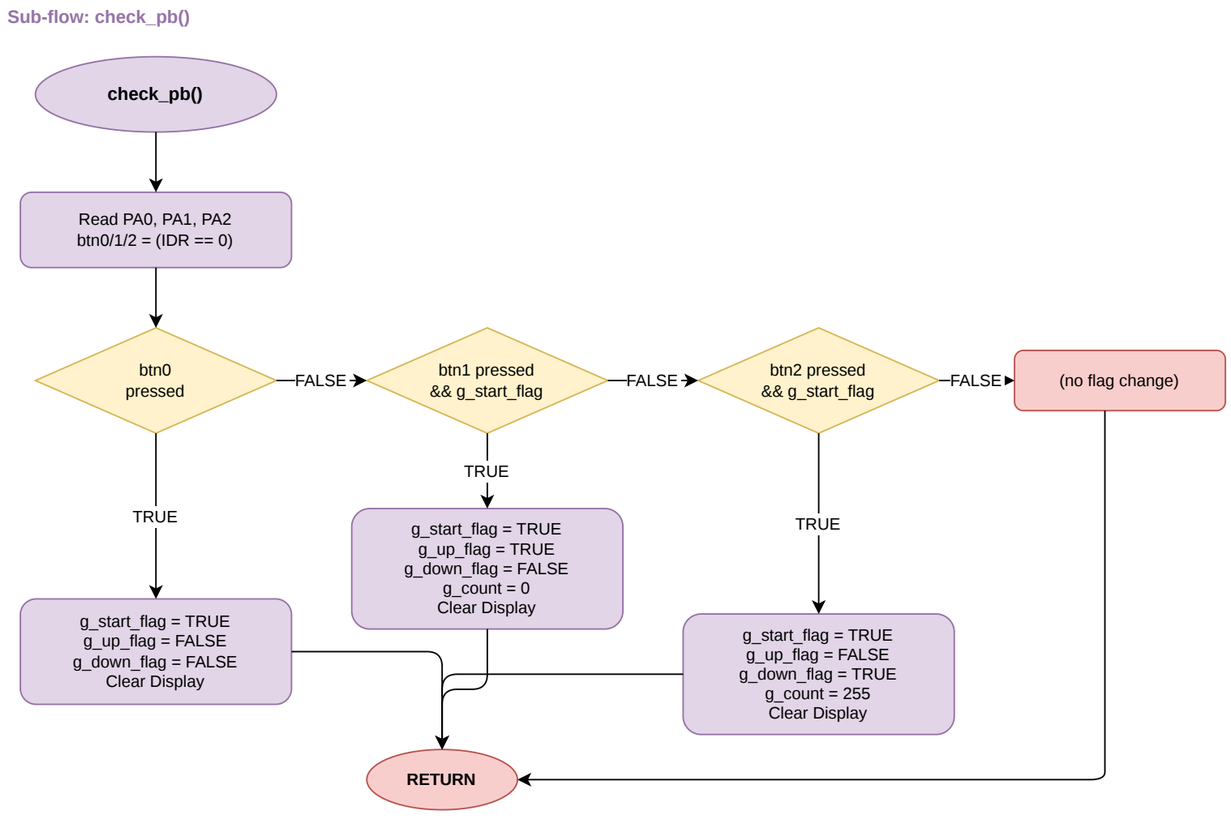
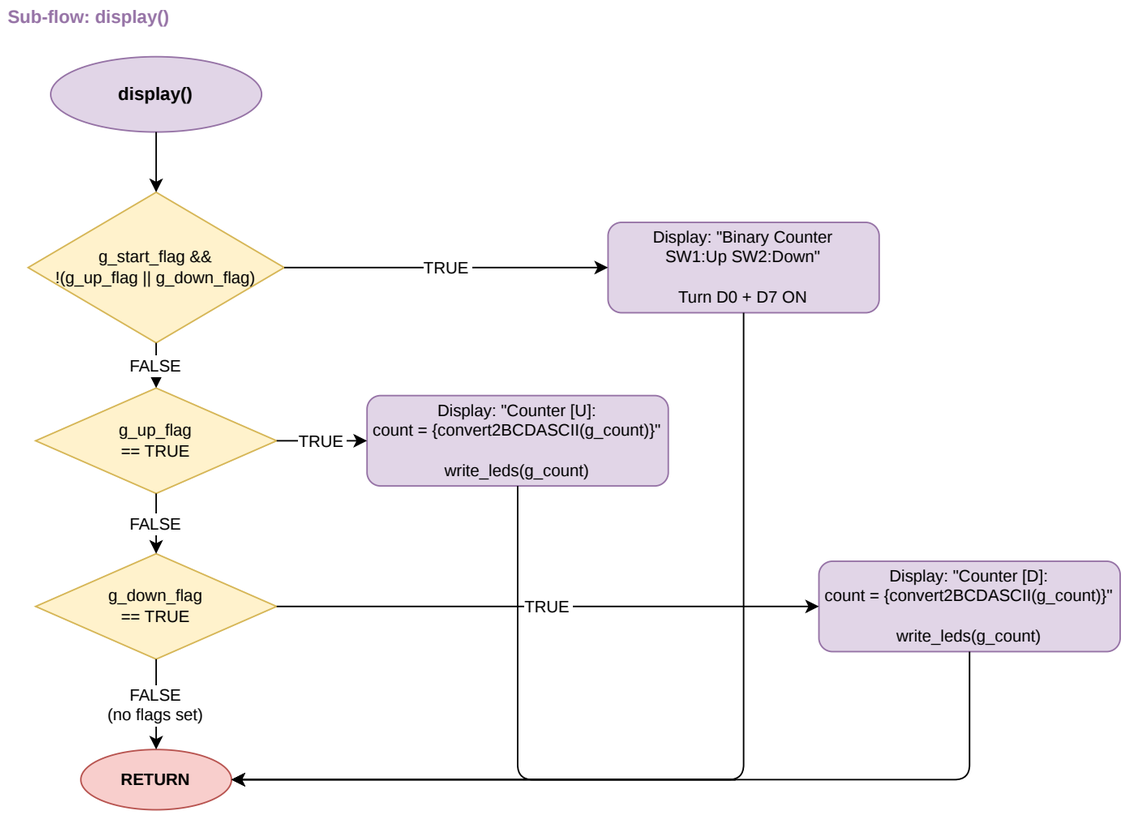
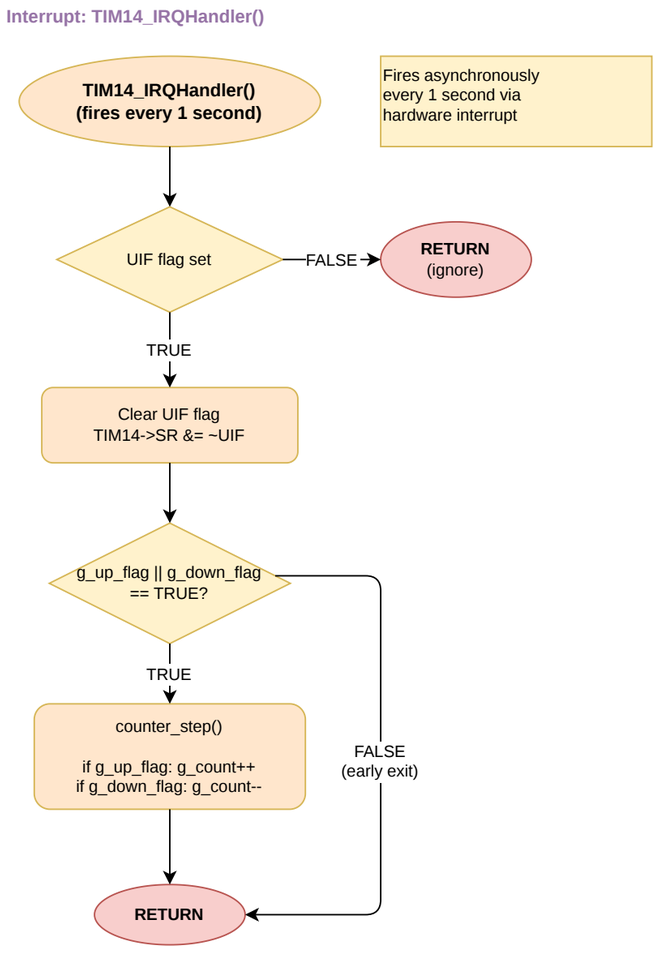

START
init_clock()
init_gpio()
PA0–PA2 = inputs
PB0–PB7 = outputs
Turn D0 ON
init_timer()
Enable Interrupt
Start timer (1s period)
LCD Init
Splash Screen
MAIN LOOP
check_pb()
Read PA0–PA2, update flags
display()
Update LCD and LEDs
based on flags


Sub-flow: check_pb()
check_pb()
Read PA0, PA1, PA2
btn0/1/2 = (IDR == 0)
btn0
pressed
FALSE
btn1 pressed
&& g_start_flag
FALSE
btn2 pressed
&& g_start_flag
FALSE
(no flag change)
TRUE
g_start_flag = TRUE
g_up_flag = FALSE
g_down_flag = FALSE
Clear Display
TRUE
g_start_flag = TRUE
g_up_flag = TRUE
g_down_flag = FALSE
g_count = 0
Clear Display
TRUE
g_start_flag = TRUE
g_up_flag = FALSE
g_down_flag = TRUE
g_count = 255
Clear Display
RETURN


Sub-flow: display()
display()
g_start_flag &&
!(g_up_flag || g_down_flag)
TRUE
Display: "Binary Counter
SW1:Up SW2:Down"
Turn D0 + D7 ON
FALSE
g_up_flag
== TRUE
TRUE
Display: "Counter [U]:
count = {convert2BCDASCII(g_count)}"
write_leds(g_count)
FALSE
g_down_flag
== TRUE
TRUE
Display: "Counter [D]:
count = {convert2BCDASCII(g_count)}"
write_leds(g_count)
FALSE
(no flags set)
RETURN


Interrupt: TIM14_IRQHandler()
TIM14_IRQHandler()
(fires every 1 second)
Fires asynchronously
every 1 second via
hardware interrupt
UIF flag set
FALSE
RETURN
(ignore)
TRUE
Clear UIF flag
TIM14->SR &= ~UIF
g_up_flag || g_down_flag
== TRUE?
TRUE
counter_step()
if g_up_flag: g_count++
if g_down_flag: g_count--
FALSE
(early exit)
RETURN